Chapter 3 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. When the following equation is balanced, the coefficients are ________.

$$NH_3(g) + O_2(g) \rightarrow NO_2(g) + H_2O(g)$$

A) 4, 3, 4, 3
B) 2, 3, 2, 3
C) 8, 14, 8, 12
D) 4, 7, 4, 6
E) 1, 3, 1, 2

2. Which of the following reactions is the balanced equation that represents the decomposition reaction that occurs when silver oxide is heated?
A) $AgO(s) \rightarrow Ag(s) + O(g)$
B) $2\ AgO(s) \rightarrow Ag(s) + O_2(g)$
C) $Ag_2O(s) \rightarrow 2\ Ag(s) + O(g)$
D) $Ag_2O(s) \rightarrow 2\ Ag(s) + O_2(g)$
E) $2\ Ag_2O(s) \rightarrow 4\ Ag(s) + O_2(g)$

3. Which of the following is the correct formula weight for calcium phosphate?
A) 310.2 amu B) 135.1 amu C) 182.2 amu D) 278.2 amu E) 175.1 amu

4. What is the percentage of nitrogen, by mass, in aluminium nitrate?
A) 15.7% B) 18.6% C) 19.7% D) 23.0% E) 80.3%

5. A sample of CH_2F_2 with a mass of 19 g contains ________ atoms of F.
A) 2.2×10^{23} B) 38 C) 3.3×10^{24} D) 4.4×10^{23} E) 0.73

6. What mass of O_2 is consumed in the complete combustion of 1.00 g of methanol, (CH_3OH)?
A) 1.50 g B) 2.00 g C) 2.50 g D) 3.00 g E) 5.00 g

7. Consider the following reaction: $2\,A + B \rightarrow 3\,C + D$
3.0 mol *A* and 2.0 mol *B* react to form 4.0 mol *C*. What is the percent yield of this reaction?
A) 50% B) 67% C) 75% D) 89% E) 100%

8. A compound whose empirical formula is XF_3 consists of 65% F by mass. What is the atomic mass of X?
A) 14 B) 80 C) 40 D) 35 E) 31

9. Sulfuric acid may be produced by the following process:

$$4\ FeS_2 + 11\ O_2 \rightarrow 2\ Fe_2O_3 + 8\ SO_2$$
$$2\ SO_2 + O_2 \rightarrow 2\ SO_3$$
$$SO_3 + H_2O \rightarrow H_2SO_4$$

How many moles of H_2SO_4 will be produced from 6.15 moles of FeS_2?
A) 33.8 B) 6.15 C) 12.3 D) 3.08 E) 1.54

10. A 2.203-g sample of a compound known to contain only carbon, hydrogen, and oxygen was burned in oxygen to produce 4.401 g CO_2 and 1.802 g H_2O. A separate experiment shows that it has a molar mass of 88.1 g/mol. What is the molecular formula of the compound?
A) C_2H_4O B) $C_4H_8O_2$ C) $C_5H_{12}O$ D) $C_4H_4O_2$ E) $C_3H_4O_3$

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Part B Short questions (Write legibly and show work for all steps in the following problems. For mathematical relationships, show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures.)

11. (2 points) Rank the following samples in order of increasing numbers of carbon atoms:

24 g ^{12}C*, 1 mol* C_3H_8*,* 9×10^{23} *molecules of carbon dioxide, 1 mol of lithium carbonate*

12. (6 points) Write the balanced chemical equations (indicating the states) for (**a**) the complete combustion of acetic acid (CH_3COOH), the main active ingredient in vinegar; (**b**) the decomposition of solid sodium nitrate into solid sodium nitrite and oxygen gas; (**c**) the combination reaction between magnesium metal and chlorine gas.

13. (7 points) When a mixture of 10.0 g of acetylene (C_2H_2) and 10.0 g of oxygen (O_2) is ignited, the resultant combustion reaction produces CO_2 and H_2O. (**a**) Which is the limiting reactant? (**b**) How many grams of C_2H_2, O_2, CO_2, and H_2O are present after the reaction is complete?

Chapter 3 Homework Answer Sheet

Name: **Student ID:** **Instructor:** **Score:**

Part A Multiple choice (10 points)

1-5______________

6-10_____________

Part B Short questions (15 points)